Hoan Lam

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SUMMARY

Software engineer focused on building real-world systems across computer vision, full-stack development, and automation. Experience developing end-to-end applications from data processing to user-facing products with an emphasis on practical impact.

EXPERIENCE

Blue Nucleus May 2025 – Present
Software Developer Grand Rapids, MI

- Developed software features and tools to support internal workflows and client-facing applications
- Collaborated with cross-functional teams to deliver functional solutions aligned with project requirements
- Improved system efficiency and usability through iterative development and testing

PROJECTS

Find Queen Bee (Capstone) – GVSU

- Built a real-time queen bee detection system using YOLO
- Developed the training data pipeline, including image labeling and preprocessing
- Improved model performance through dataset refinement and model tuning, then deployed it into a Flutter mobile app for iOS and Android

Valent Calculator (Chemical Application Tool) – Work

- Developed a cross-platform mobile application for agricultural chemical conversions and dosing calculations
- Built the app using React Native, Expo, and Zustand, with conversion logic for volume, weight, and concentration
- Integrated Firebase for product data management and set up CI/CD pipelines using EAS

Learning Companion Web App – GVSU Research

- Designed and developed a web platform for teachers and students with integrated AI features
- Built the application using Next.js, Zustand, ShadCN UI, and Supabase
- Developed features to collect and analyze student interaction data to identify unusual learning patterns in K–12 settings

Lacrosse Dashboard (Digital Sign System) – Work

- Built a real-time dashboard using React to display live game statistics and scores
- Converted the web application into a desktop app using PyWebView based on customer requirements
- Implemented serial communication between the dashboard and microcontroller to enable real-time data updates

3D File Management System (Engineering Lab Tool) – Personal

- Developed a desktop application using Electron with a React-based UI to manage 3D printing workflows across queue, in-progress, and completed stages
- Implemented drag-and-drop file handling and metadata tracking using SQLite for efficient data management
- Designed system to support lab operations and inventory tracking through structured file organization

EDUCATION

Grand Valley State University Expected May 2026
Master of Science in Applied Computer Science GPA: 3.9

SKILLS

Languages: Python, JavaScript, TypeScript, Dart, Kotlin
Frameworks: React, React Native, Flask, Next, Vue, Flutter
Tools: Git, Docker, Firebase, Supabase, EAS, Apple Connect, Kestra
Other: OpenCV, YOLO, Segment Anything (SAM), REST APIs